

Townsend: 10.3, 10.7, 10.8, 10.19, 11.1, 11.4

Townsend Problem 10.3 (5 pts)

A negatively charged pion (a spin-0 particle) is bound to a proton forming a pionic hydrogen atom. At time $t = 0$ the system is in the state

$$|\psi\rangle = \frac{1}{2}|1,0,0\rangle + \frac{1}{\sqrt{2}}|2,1,1\rangle + \frac{1}{2}|2,1,0\rangle$$

In an external magnetic field in the z direction, the Hamiltonian is given by

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{2\mu} - \frac{e^2}{|\hat{\mathbf{r}}|} + \omega_0 \hat{L}_z$$

where μ is the reduced mass of the pion-proton system.

- (a) What is $|\psi(t)\rangle$ for the system at time t ? What is $\langle E \rangle$ at time t for this state?
- (b) What is $\langle L_z \rangle$ and $\langle L_x \rangle$ at time t for this state?

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Townsend Problem 10.7 (5 pts)

An electron is in the ground state of tritium, for which the nucleus is the isotope of hydrogen with one proton and two neutrons. A nuclear reaction instantaneously changes the nucleus into ${}^3\text{He}$, which consists of two protons and one neutron. Calculate the probability that the electron remains in the ground state of the new atom. Obtain a numerical answer.

Note: It is sufficient to treat the mass of the proton and neutron as equal.

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Townsend Problem 10.8 (5 pts)

Show that there are no allowed energies $E < -V_0$ for the potential well

$$V = \begin{cases} -V_0 & r < a \\ 0 & r > a \end{cases}$$

by explicitly solving the Schrödinger equation and attempting to satisfy all the appropriate boundary conditions.

Note: $\ell = 0$ corresponds to the ground state. Start by solving the radial equation for $u(r) = rR(r)$ with $\ell = 0$ for $r < a$ and $r > a$ and then, as the problem indicates, attempting to satisfy the appropriate boundary conditions, namely at $r = 0$, $r = a$, and r approaching infinity.

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Townsend Problem 10.19 (5 pts)

The Hamiltonian for two spin- $\frac{1}{2}$ particles, one with mass m_1 and the other with mass m_2 , is given by

$$\hat{H} = \frac{\hat{\mathbf{p}}_1^2}{2m_1} + \frac{\hat{\mathbf{p}}_2^2}{2m_2} + V_a(|\hat{\mathbf{r}}|) + \left(\frac{1}{4} - \frac{\hat{\mathbf{S}}_1 \cdot \hat{\mathbf{S}}_2}{\hbar^2} \right) V_b(|\hat{\mathbf{r}}|)$$

where $\hat{\mathbf{r}} = \hat{\mathbf{r}}_1 - \hat{\mathbf{r}}_2$ and

$$V_a(r) = \begin{cases} 0 & r < a \\ V_0 & r > a \end{cases} \quad V_b(r) = \begin{cases} 0 & r < b \\ V_0 & r > b \end{cases}$$

with $b < a$ and V_0 very large and positive.

- (a) Determine the normalized position-space energy eigenfunction for the ground state. What is the spin state of the ground state? What is the degeneracy? *Note:* Take V_0 to be infinite.
- (b) What can you say about the energy and spin state of the first-excited state? Does your result depend on how much larger a is than b ? Explain.

Note: What are the eigen states of the spin-dependent term in parenthesis? Each of these spin configurations can have its own position-space wave function associated with it.

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Townsend Problem 11.1 (5 pts)

Consider a perturbation $\hat{H}_1 = b\hat{x}^4$ to the simple harmonic oscillator Hamiltonian

$$\hat{H}_0 = \frac{\hat{p}_x^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2$$

This is an example of an anharmonic oscillator, one with a nonlinear restoring force.

- (a) Show that the first-order shift in the energy is given by

$$E_n^{(1)} = \frac{3\hbar^2 b}{4m^2\omega^2}(1 + 2n + 2n^2)$$

- (b) Argue that no matter how small b is, the perturbation expansion will break down for some sufficiently large n . What is the physical reason?

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Townsend Problem 11.4 (5 pts)

Calculate the first-order shift to the energy of the ground state and first-excited state of a particle of mass m in the one-dimensional infinite square well

$$V(x) = \begin{cases} 0 & 0 < x < L \\ \infty & \text{elsewhere} \end{cases}$$

of (a) the constant perturbation $\hat{H}_1 = V_1$ and (b) the linearly increasing perturbation $\hat{H}_1 = \varepsilon E_1^{(0)} \hat{x}/L$, where $E_1^{(0)}$ is the unperturbed energy of the ground state and $\varepsilon \ll 1$.

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